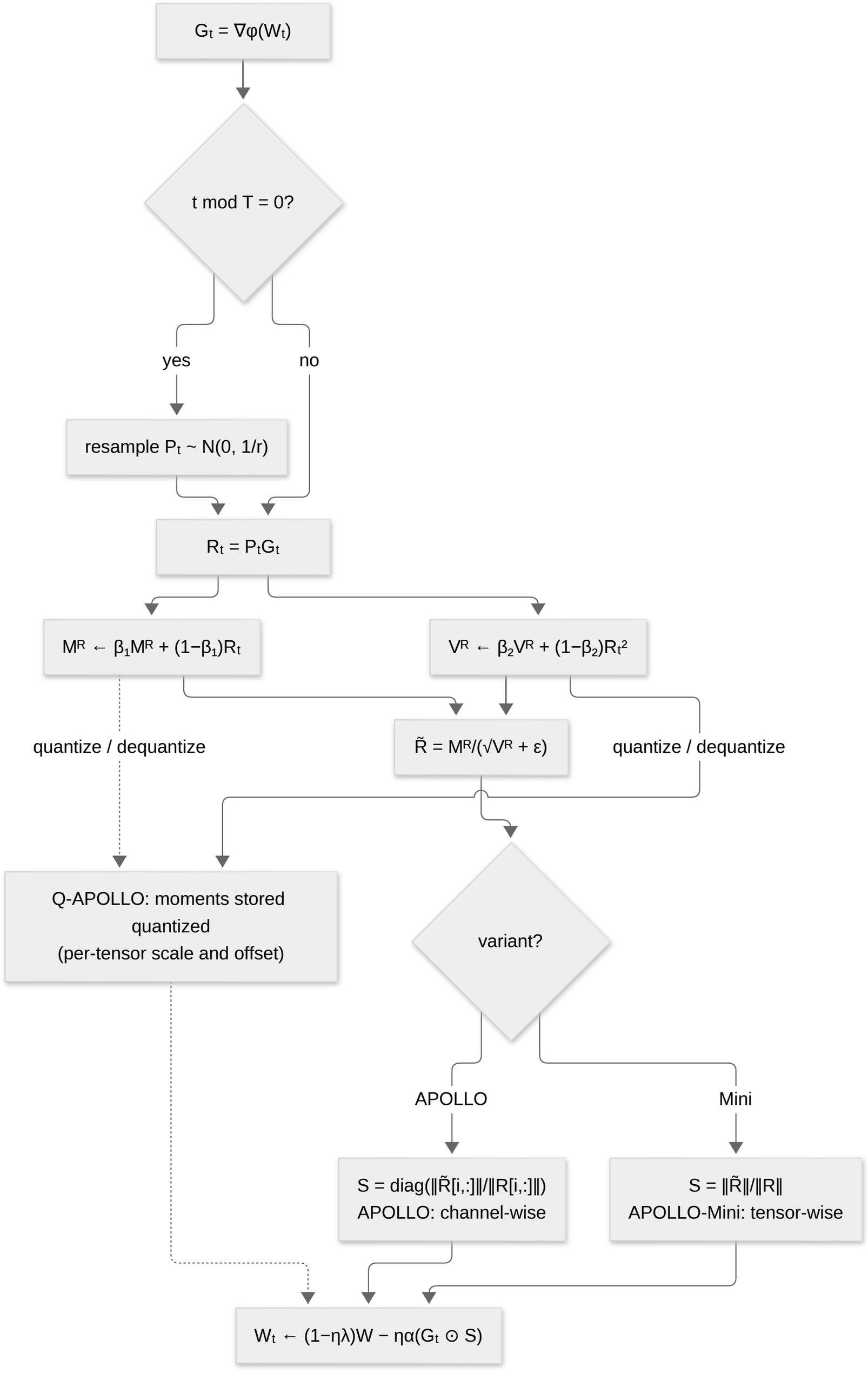

$G_t = \nabla\varphi(W_t)$
$t \bmod T = 0$?
yes
no
resample $P_t \sim N(0, 1/r)$
$R_t = P_t G_t$
$M^R \leftarrow \beta_1 M^R + (1-\beta_1) R_t$
$V^R \leftarrow \beta_2 V^R + (1-\beta_2) R_t^2$
quantize / dequantize
$\tilde{R} = M^R/(\sqrt{V^R} + \varepsilon)$
quantize / dequantize
Q-APOLLO: moments stored quantized (per-tensor scale and offset)
variant?
APOLLO
Mini
$S = \mathrm{diag}(\|\tilde{R}[i,:]\|/\|R[i,:]\|)$ APOLLO: channel-wise
$S = \|\tilde{R}\|/\|R\|$ APOLLO-Mini: tensor-wise
$W_t \leftarrow (1-\eta\lambda)W - \eta\alpha(G_t \odot S)$